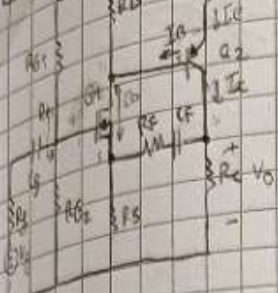


Homework 2

EX 1



$$V_G = V_{CC} R_{G2} / (R_{G1} + R_{G2}) = 4.5V$$

$$V_{GS} = V_G - I_D R_S \Rightarrow I_D = (V_G - V_{GS}) / R_S$$

$$I_D = I_{DSS} (1 - V_{GS} / V_t)^2 = (V_G - V_{GS})^2 / R_S$$

$$1 + \frac{V_{GS}^2}{V_t^2} - \frac{2V_{GS}}{V_t} = \frac{V_G}{R_S I_{DSS}} - \frac{V_{GS}}{R_S I_{DSS}}$$

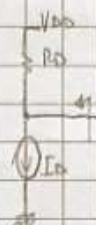
$$\frac{V_{GS}^2}{V_t^2} + V_{GS} \left(\frac{1}{R_S I_{DSS}} - \frac{2}{V_t} \right) + 1 - \frac{V_G}{R_S I_{DSS}} = 0$$

$$V_{GS} = 4.16V > V_t$$

$$I_D = 0.336mA$$

$$I_C = 0.52mA$$

$$V_{CE} = - (V_{DD} - R_E I_C (1 + \frac{1}{\beta})) - I_C R_C = -2.64V \approx V_{CE}$$



$$(I_D - I_E) R_D = I_E (1 + \frac{1}{\beta}) R_E - V_{GS}$$

$$I_C = \frac{I_D R_D + V_{GS}}{\frac{R_D}{\beta} + R_E (1 + \frac{1}{\beta})} = 0.52mA$$

Small signal analysis:

feedback topology: voltage sampling, voltage sampling → voltage amplifier

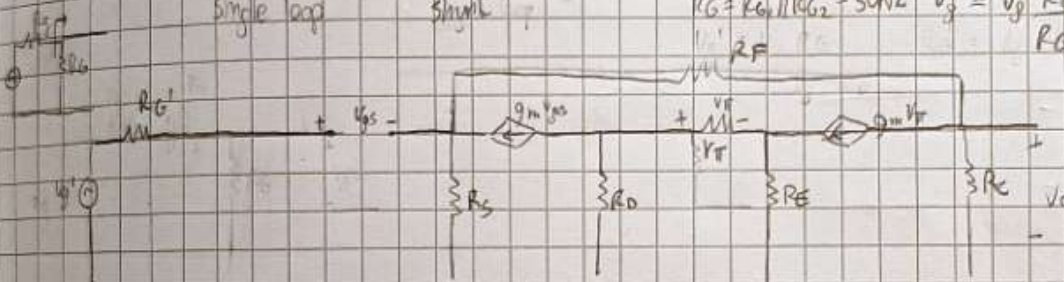
Single loop

Shunt

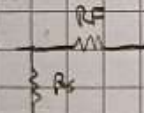
$$R_G = R_{G1} || R_{G2} = 50k\Omega$$

$$V_g' = V_g \frac{R_g}{R_g + R_g} = 0.5V$$

$$R_g' = R_g || R_g = 25k\Omega$$



betweenthink



$$R_{if} = R_S || R_F = 952\Omega$$

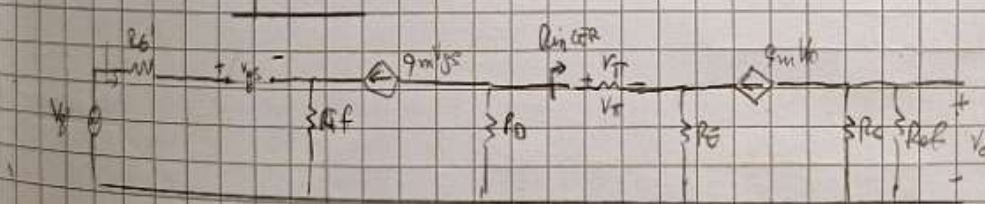
$$R_{of} = R_S + R_F = 21k\Omega$$

$$\beta = \frac{R_S}{R_S + R_F} = 0.0476$$

$$g_{m_{BJT}} = \frac{2I_D}{V_{GS} - V_t} = 5.29 \cdot 10^{-4} S$$

$$g_{m_{BJT}} = \frac{I_C}{V_t} = 0.0208 \cdot 10^{-3} S$$

$$r_f = \frac{R_D}{g_{m_{BJT}}} = 6.808k\Omega$$



$$A_{CE} = -\beta R_{CE} R_{of} = -2.984$$

$$A_v = \frac{V_o}{V_{g'}} = \alpha \cdot \alpha_{CE} = 6.24$$

$$R_{CE} = V_{CC} + (1 + \beta) R_E = 227k\Omega$$

$$A_F = \frac{A_v}{1 + \beta A_v} = 4.81$$

$$\alpha_{CE} = -\frac{g_m R_D}{1 + g_m R_E} = -2.09$$

$$A_v = \frac{V_o}{V_g} = \frac{V_o}{V_{g'}} \cdot \frac{V_{g'}}{V_g} = A_F \cdot \frac{R_D}{R_D + R_g} = 4.81$$

$$\beta = 1$$

Pin is not modified by feedback

$$P_{in} = R_G = R_{G1} || R_{G2} = 50k\Omega$$

Noting RF

$$2\pi R_g C_f = 312Hz$$

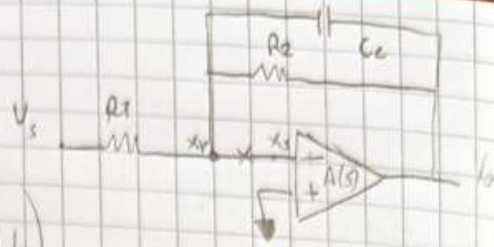
$$R_{g'} = R_g + R_G = 51k\Omega$$

Ex 2

$$R_1 = 1k\Omega$$

$$R_2 = 33k\Omega$$

$$A(s) = \frac{A_0}{(1 + \frac{s}{\omega_{p1}}) (1 + \frac{s}{\omega_{p2}})}$$



$$1) T = \frac{V_o}{V_s} = \frac{R_1 / (R_1 + R_2 \parallel \frac{1}{sC_e})}{1} V_o = A(s) \cdot \left(\frac{R_1}{R_1 + R_2 \parallel \frac{1}{sC_e}} \right)$$

If we don't consider C_e $\beta = \frac{R_1}{R_1 + R_2} = 0.0294$

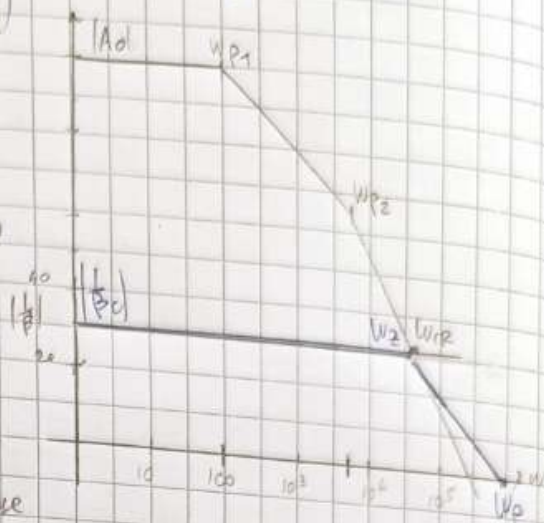
$$\frac{1}{\beta} = \frac{R_1 + R_2}{R_1} = 36 \Rightarrow 31dB \quad (34 > 10 \text{ we can have more than } 45^\circ \Delta\phi)$$

$$A_0 \omega_{p1} = A_1 \omega_{p2}$$

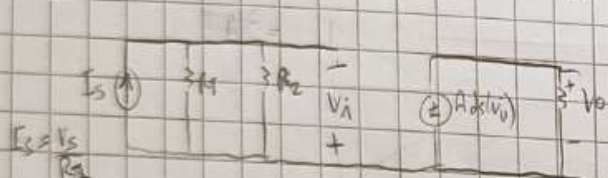
$$A_1 \omega_{p2}^2 = \frac{1}{\beta} \omega_{cr}^2 \Rightarrow \omega_{cr} = \sqrt{\beta \omega_{p1} \omega_{p2} A_0} = 38.35 \text{ Krad/s}$$

$$PM = 180 - \arctan\left(\frac{\omega_{cr}}{\omega_{p1}}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_{p2}}\right) = 7.58^\circ$$

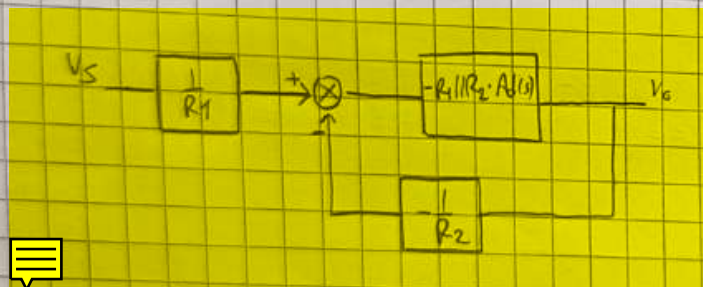
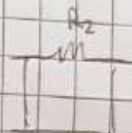
low phase margin



2) From feedback theory: current mixing and voltage sampling $\rightarrow \frac{V_o}{V_s} \Rightarrow$ Transresistance amplifier



$$\beta = -\frac{1}{R_2} = \frac{V_i}{V_o} \bigg|_{V_o=0}$$



$$3) \beta_c = \frac{R_1}{R_1 + R_2} = \frac{R_1}{R_1 + R_2} \frac{1 + sR_2C_e}{1 + s \frac{R_1R_2}{R_1 + R_2} C_e} = \beta_0 \frac{1 + s/\omega_z}{1 + s/\omega_p}$$

$$\omega_z = \omega_{cr} = \frac{1}{R_2 C_e} \Rightarrow C_e = \frac{1}{R_2 \omega_{cr}} = 0.79 \text{ nF}$$

$$\beta_{cr} = 1 = 0dB$$

$$\omega_z = \omega_{cr}$$

$$\omega_p = \frac{R_1 + R_2}{R_2 R_1 C_e} = 1.306 \text{ Mrad/s}$$

low pass filter bandwidth $\omega_z = \omega_{cr}$ (expect 45° Δφ)

$$PM = 180 - \arctan\left(\frac{\omega_{cr}}{\omega_{p1}}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_{p2}}\right) + \arctan\left(\frac{\omega_{cr}}{\omega_z}\right) - \arctan\left(\frac{\omega_{cr}}{\omega_p}\right) = 50.9^\circ$$

we increased of $\Delta\phi = 50.9^\circ - 7.58^\circ = 43.3^\circ$

1 and e Smulth